

Week 06: Except, IO, and do

Functional Programming in Lean

Learning goals

- ▶ Explicit error types
- ▶ Monadic sequencing with `do`
- ▶ IO actions

Core concepts

- ▶ `Except String Nat`: success or error
- ▶ `do`: sequencing and error propagation
- ▶ `I0`: interaction with the external world

Lean example

```
def safeDivide (x y : Nat) :  
  Except String Nat :=  
  if y == 0 then .error "division by zero"  
  else .ok (x / y)
```

Lean details

- ▶ `.ok` and `.error` constructors
- ▶ `←` for a monadic result
- ▶ Error handling in the return type

Live coding

- ▶ `safeDivide`
- ▶ `ratioOfSums` in `do` notation
- ▶ Success and failure with `#eval`

Check

Result of ratioOfSums 10 2 0 0

Exercise

- ▶ `divideAfterIncrement`
- ▶ Explicit match versus do